

Introduction to Cyber Security

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Introduction

- Cyber security:
 - All about reducing risk of attacks to computers, networks, or software.
 - **Malicious actors (offense)**: try to attack assets such as websites or company networks.
 - **Cyber security safeguards (defence)**: are put in place to stop these attacks.
 - Offense is always way ahead of defence:
 - Offense must only find one successful path to the targeted asset; defence must prevent all such paths.

Security Services

- Cyber security rests on **confidentiality**, **integrity** and **availability** of resources:
 - Confidentiality: concealment of information or resources
 - Access control mechanisms support confidentiality for example encryption
 - Integrity
 - Trustworthiness of data or resources and is usually phrased in terms of preventing improper or unauthorized change.
 - Includes:
 - Data integrity: content
 - Origin integrity: source of the data; also called authentication

Security Services cont.

- Integrity mechanism fall into 2 classes:
 - **Prevention mechanisms**: block any unauthorized attempts to change the data or any attempts to change the data in unauthorized ways
 - **Detection mechanisms**: report that the data's integrity is no longer trustworthy.
- Availability
 - The ability to use the information or resource desired.
 - Attempts to block availability, called denial of service attacks, can be most difficult to detect because the analyst must determine if the unusual access patterns are attributable to deliberate manipulation of resources or of environment.

Threats

- A potential violation of security.
 - The violation need not actually occur for there to be a threat.
- Actions that may cause a violation to occur are called **attacks**.
- Threats can be divided into 4 broad classes:
 - **Disclosure**, or unauthorized access to information
 - **Deception**, or acceptance of false data.
 - **Disruption**, or interruption or prevention of correct operation
 - **Usurpation**, or unauthorized control of some part of a system.
 - **Modification** or alteration: unauthorized change of information.

In which category can you place each of the following?

- Snooping
 - Snooping, the unauthorized interception of information
- Masquerading or spoofing
 - Impersonation of one entity by another. Authentication services counter this threat.
- Repudiation of origin
 - A false denial that an entity created or sent something.
- Denial of receipt
 - A false denial that an entity received some information
- Delay
 - A temporary inhibition of service

In which category can you place each of the following?

- Denial of service
 - A long-term inhibition of service. The attacker prevents a server from providing some service

Policy and mechanism

- A security policy is a statement of **what is**, and **what is not**, allowed.
- A security mechanism is a method, tool, or procedure for enforcing a security policy.

Goals of Security

- Prevention
 - An attack fails
- Detection
 - To determine that an attack is underway, or has occurred, and to report it.
- Recovery
 - Stop an attack and assess and recover any damage caused by that attack.
 - Identification and fixing of the vulnerability that led to the attack.

Cyber Attacks

- Attacks exploit vulnerabilities.
- 2 forms:
 - Brute-force attacks
 - Heuristic attack
- Hacking campaign combines these techniques.
- When nation states perform these attacks over a long period of time, we call this an advanced persistent attack, or advanced persistent threat (APT)
- Example attack: **SQL injection**

Malicious Actors

- Who is doing the hacking, with what motivation?
 - Hackers:
 - White hats hack to help owners
 - Black hats hack to embarrass owners
 - Grey hats are in between
 - Cyber criminals: motivated by money. Often rely on fraudulent use of stolen accounts and are frequently found targeting any thing with financial value such as credit cards, medical records, etc, that can be sold on the Dark Web.
 - Cyber terrorists: driven by some political or philosophical motivation.
 - Nation state attackers

Malicious actors cont.

- Nation state attackers often:
 - Use APTs on industrial targets to steal intellectual property
 - Use cyber weapons to disrupt, break or deny access to an adversary's critical infrastructure.

Cyber defenders

- Individuals, for example against phishing attacks.
- Enterprise security teams
- Cyber security technology vendors
- Government and regulatory organizations: try to reduce risk through legal, policy, and oversight methods.
- Cyber military and intelligence organizations.

References and Further Reading

- From CIA to APT: An introduction to cyber security. Edward Amoroso. Matthew Amoroso.
- Introduction to computer security. Matt Bishop.